Absolutely — here's a **"more than enough" syllabus checklist** for **Pandas + Matplotlib + Seaborn**, designed to take you from beginner to strong practical/data-analysis level.

🐼 Pandas — Complete Checklist

1. Basics

- ☐ What is Pandas?
- ☐ Installation/importing Pandas
- ☐ `Series`
- ☐ `DataFrame`
- ☐ Creating Series/DataFrames
- ☐ `index`
- ☐ `columns`
- ☐ `dtypes`
- ☐ `shape`
- ☐ `size`
- ☐ `ndim`
- ☐ `values`
- ☐ `info()`
- ☐ `describe()`
- ☐ `head()` / `tail()`

2. Reading & Writing Data

- ☐ `read_csv()`
- ☐ `read_excel()`
- ☐ `read_json()`
- ☐ `read_sql()`
- ☐ Reading from URLs
- ☐ `to_csv()`
- ☐ `to_excel()`
- ☐ `to_json()`
- ☐ `to_sql()`
- ☐ Understanding file paths

- [ ] `parse_dates`
- [ ] Handling encoding

3. Selecting Data

- [ ] Selecting columns
- [ ] Selecting multiple columns
- [ ] Selecting rows
- [ ] `loc[]`
- [ ] `iloc[]`
- [ ] Boolean filtering
- [ ] Multiple conditions
- [ ] `isin()`
- [ ] `between()`
- [ ] `query()`
- [ ] `at[]`
- [ ] `iat[]`

4. Data Cleaning

- [ ] Detecting missing values
- [ ] `isna()`
- [ ] `isnull()`
- [ ] `notna()`
- [ ] `dropna()`
- [ ] `fillna()`
- [ ] Forward fill
- [ ] Backward fill
- [ ] Missing-value strategies
- [ ] Detecting duplicates
- [ ] `duplicated()`
- [ ] `drop_duplicates()`
- [ ] Replacing values
- [ ] `replace()`
- [ ] Fixing inconsistent values
- [ ] Handling incorrect data types

5. Data Types

- [ ] `astype()`
- [ ] Numeric conversion
- [ ] `pd.to_numeric()`
- [ ] String dtype
- [ ] Category dtype
- [ ] Datetime dtype
- [ ] Boolean dtype
- [ ] Understanding `object`

6. String Operations

- [ ] `.str.lower()`
- [ ] `.str.upper()`
- [ ] `.str.title()`
- [ ] `.str.strip()`
- [ ] `.str.replace()`
- [ ] `.str.contains()`
- [ ] `.str.startswith()`
- [ ] `.str.endswith()`
- [ ] `.str.split()`
- [ ] `.str.extract()`
- [ ] Regular expressions with `.str`

7. Sorting & Ranking

- [ ] `sort_values()`
- [ ] `sort_index()`
- [ ] Multiple-column sorting
- [ ] Ascending/descending sorting
- [ ] `rank()`
- [ ] `nlargest()`
- [ ] `nsmallest()`

8. Indexing

- [ ] Setting index
- [ ] `set_index()`
- [ ] `reset_index()`
- [ ] Renaming index
- [ ] Renaming columns
- [ ] MultiIndex basics
- [ ] Index alignment
- [ ] Reindexing

9. Aggregation

- [ ] `sum()`
- [ ] `mean()`
- [ ] `median()`
- [ ] `mode()`
- [ ] `min()` / `max()`
- [ ] `count()`
- [ ] `std()`
- [ ] `var()`
- [ ] `quantile()`
- [ ] `nunique()`
- [ ] `value_counts()`
- [ ] `agg()`

10. GroupBy — ⭐ Very Important

- [ ] `groupby()`
- [ ] Grouping by one column
- [ ] Grouping by multiple columns
- [ ] Aggregating groups
- [ ] Multiple aggregations
- [ ] Named aggregation
- [ ] `transform()`
- [ ] `filter()`
- [ ] `apply()`
- [ ] GroupBy + sorting
- [ ] GroupBy + `value_counts()`

11. Combining Data

- [ ] `concat()`
- [ ] `merge()`
- [ ] Inner join
- [ ] Left join
- [ ] Right join
- [ ] Outer join
- [ ] Cross join
- [ ] Joining on multiple columns
- [ ] `join()`
- [ ] Understanding primary/foreign-key-like relationships

12. Reshaping

- [ ] `pivot()`
- [ ] `pivot_table()`
- [ ] `melt()`
- [ ] `stack()`
- [ ] `unstack()`
- [ ] `transpose()`
- [ ] Wide vs long data

13. Datetime

- [ ] `pd.to_datetime()`
- [ ] `.dt.year`
- [ ] `.dt.month`
- [ ] `.dt.day`
- [ ] `.dt.hour`
- [ ] `.dt.day_name()`
- [ ] `.dt.month_name()`
- [ ] Date filtering
- [ ] Date differences
- [ ] `Timedelta`
- [ ] Date ranges
- [ ] Resampling

- [ ] Rolling windows

14. Advanced Pandas

- [ ] `apply()`
- [ ] `map()`
- [ ] `applymap()` / element-wise alternatives
- [ ] Lambda functions
- [ ] `transform()`
- [ ] Window functions
- [ ] `rolling()`
- [ ] `expanding()`
- [ ] `ewm()`
- [ ] Categorical data
- [ ] MultiIndex
- [ ] Memory optimization
- [ ] Efficient filtering
- [ ] Vectorization
- [ ] Avoiding unnecessary loops
- [ ] Copy vs view
- [ ] `SettingWithCopy` issues
- [ ] Method chaining

📊 Matplotlib — Complete Checklist

1. Fundamentals

- [ ] What is Matplotlib?
- [ ] `pyplot`
- [ ] `figure()`
- [ ] `show()`
- [ ] Figure vs Axes
- [ ] Creating subplots
- [ ] `subplots()`

2. Basic Charts

- [ ] Line plot
- [ ] Bar chart
- [ ] Horizontal bar chart
- [ ] Scatter plot
- [ ] Histogram
- [ ] Pie chart
- [ ] Area plot
- [ ] Stem plot
- [ ] Box plot
- [ ] Violin plot
- [ ] Error bars

3. Customization

- [ ] Figure size
- [ ] Colors
- [ ] Line styles
- [ ] Line width
- [ ] Markers
- [ ] Marker size
- [ ] Transparency (`alpha`)
- [ ] Labels
- [ ] Titles
- [ ] Legends
- [ ] Grid
- [ ] Axis limits
- [ ] Tick labels
- [ ] Tick rotation
- [ ] Fonts

4. Subplots

- [ ] Multiple plots
- [ ] Rows and columns
- [ ] `fig, ax`
- [ ] `ax.plot()`
- [ ] `ax.bar()`
- [ ] `ax.scatter()`
- [ ] Shared axes

- [ ] `tight_layout()`
- [ ] `constrained_layout`

5. Advanced Visualization

- [ ] Multiple lines
- [ ] Multiple axes
- [ ] Secondary y-axis
- [ ] Annotations
- [ ] Text on plots
- [ ] Horizontal/vertical reference lines
- [ ] Logarithmic scales
- [ ] Date plots
- [ ] Colormaps
- [ ] Colorbars
- [ ] Custom ticks
- [ ] Custom legends

6. Saving

- [ ] `savefig()`
- [ ] PNG
- [ ] JPG
- [ ] SVG
- [ ] PDF
- [ ] DPI
- [ ] Transparent backgrounds
- [ ] Bounding box

🎨 Seaborn — Complete Checklist

1. Fundamentals

- [ ] What is Seaborn?
- [ ] Seaborn vs Matplotlib
- [ ] Loading datasets
- [ ] Figure-level vs axes-level functions

- [ ] Themes
- [ ] Styles
- [ ] Color palettes

2. Relational Plots

- [ ] `scatterplot()`
- [ ] `lineplot()`
- [ ] `relplot()`
- [ ] `hue`
- [ ] `style`
- [ ] `size`
- [ ] `col`
- [ ] `row`

3. Distribution Plots

- [ ] `histplot()`
- [ ] `kdeplot()`
- [ ] `displot()`
- [ ] `ecdfplot()`
- [ ] Histograms
- [ ] KDE
- [ ] Bins
- [ ] Density
- [ ] Multiple distributions

4. Categorical Plots

- [ ] `barplot()`
- [ ] `countplot()`
- [ ] `boxplot()`
- [ ] `violinplot()`
- [ ] `stripplot()`
- [ ] `swarmplot()`
- [ ] `pointplot()`
- [ ] `catplot()`

5. Regression

- [ ] `regplot()`
- [ ] `lmplot()`
- [ ] Regression line
- [ ] Confidence intervals
- [ ] Linear relationships

6. Matrix Plots

- [ ] `heatmap()`
- [ ] Correlation matrix
- [ ] `clustermap()`
- [ ] Annotations
- [ ] Color scales
- [ ] Masking

7. Pairwise Analysis

- [ ] `pairplot()`
- [ ] `PairGrid`
- [ ] `FacetGrid`
- [ ] Faceting
- [ ] Comparing multiple variables

8. Styling

- [ ] `set_theme()`
- [ ] `set_style()`
- [ ] `set_context()`
- [ ] `set_palette()`
- [ ] Built-in palettes
- [ ] Custom palettes
- [ ] Color theory for visualization

🔥 Must-Know Data Analysis Skills

Don't just learn the functions. You should be able to take a **raw dataset → insights → visualizations**.

- ☐ Load a messy CSV
- ☐ Inspect the dataset
- ☐ Identify missing values
- ☐ Remove/fill missing values
- ☐ Remove duplicates
- ☐ Fix data types
- ☐ Clean strings
- ☐ Create new columns
- ☐ Filter records
- ☐ Sort data
- ☐ Group data
- ☐ Calculate KPIs
- ☐ Merge multiple datasets
- ☐ Reshape data
- ☐ Analyze dates
- ☐ Find outliers
- ☐ Calculate correlations
- ☐ Create meaningful visualizations
- ☐ Interpret charts
- ☐ Write conclusions from data

🧠 Projects to Complete

If you can complete these without following a tutorial, your Pandas/Matplotlib/Seaborn skills will be **very solid**:

1. ☐ **Netflix dataset analysis**

- Cleaning
- Genres
- Countries
- Release years
- Ratings
- Visualizations

2. ☐ **E-commerce sales analysis**

- Monthly revenue
- Top products
- Top customers
- Category performance
- Regional performance

3. ☐ **Titanic analysis**
   - Missing values
   - Survival rate
   - Gender/age analysis
   - Class analysis
   - Correlations
4. ☐ **IPL/cricket data analysis**
   - Player performance
   - Team performance
   - Toss analysis
   - Venue analysis
   - Season trends
5. ☐ **COVID/time-series dataset**
   - Date handling
   - Rolling averages
   - Trends
   - Country comparisons
6. ☐ **Employee/HR analytics**
   - Salary analysis
   - Department comparison
   - Attrition
   - Age distribution
   - Correlation analysis
7. ☐ **Build a complete EDA**
   - Business question
   - Data cleaning
   - Univariate analysis
   - Bivariate analysis
   - Multivariate analysis
   - Visualizations
   - Findings
   - Final recommendations

⭐ Priority Order

If you don't want to learn everything at once, use this order:

Pandas

> DataFrame → indexing → filtering → cleaning → `groupby` → aggregation → merge → datetime → pivot/melt → apply/transform → advanced

Matplotlib

> Figure/Axes → line → bar → scatter → histogram → subplots → customization → annotations → saving

Seaborn

> `scatterplot` → `histplot` → `boxplot` → `barplot` → `countplot` → `violinplot` → `pairplot` → `heatmap` → `regplot` → faceting

🎯 "I'm actually good at Pandas/visualization" benchmark

You should be able to receive an unfamiliar CSV and, **without Googling every step**, do:

Load → Inspect → Clean → Transform → Analyze → Group → Merge → Visualize → Explain insights.

If you can do that across **3–5 real datasets**, you have gone well beyond the basic Pandas/Matplotlib/Seaborn syllabus.